

Shreyas Kulkarni

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EDUCATION

Masters in Computer Science

University of Southern California • Los Angeles, CA • 2023 - 2025 • **GPA:** 3.62 / 4

Coursework: Web Technologies, Operating Systems, Machine Learning, Computer Networks, Game Engine Development.

Bachelors in Computer Engineering

Pune Institute of Computer Technology • Pune, IN • 2018 - 2022 • **GPA:** 4 / 4

EXPERIENCE

Software Engineer

SARA Technology Inc.

June 2025 - Present, Los Angeles, CA

- Improved speech transcription accuracy to 90% by modifying the Wav2Vec2 model to tailor phoneme detection.
- Spearheaded state-machine game flow, reduced bug reports by 60%, streamlined transitions across 10+ levels, and tested using Jest.
- Engineered pipelines for 5K+ children's voice recordings with less than 100 ms latency using AWS Lambdas, leading a team of 3 engineers.
- Reduced children statistics computation time from 5s to 30ms by utilising DynamoDB stream triggers.
- **Skills:** React, Tailwind, Vitest, Wav2Vec2, GraphQL, AWS, DynamoDB, Python.

Backend Engineer Co-op

Fabric

February 2025 - May 2025, Los Angeles, CA

- Processed 100K+ telemetry records per day by architecting real-time analytics pipelines using AWS Step Functions.
- Fetched bitcoin related data, achieving sub-1 ms real-time data relay, integrating Twitter Activity API with Dockerized AWS Lambdas
- Minimised deployment time by 60% by writing GitHub Actions for Nuxt & CloudFormation CI/CD pipelines, checking build and code quality.
- **Skills:** AWS, Python, CI/CD, DevOps, Docker, Kubernetes.

Full Stack Engineer Intern

SARA Technology Inc.

June 2024 - August 2024, Los Angeles, CA

- Engineered React Native app with 99.8% crash-free sessions for speech therapy, partnering with designers and QA.
- Cut TAT by 50% & significantly improved delivery accuracy, streamlining over 1K+ weekly media interactions
- Improved team feature delivery velocity by 40% by architecting a scalable Recoil state management solution across 15+ screens.
- **Skills:** React, Recoil, Agile, OAuth, Swagger, Spark.

Software Engineer

Bajaj Finserv

July 2022 - June 2023, Pune, IN

- Led \$500K revenue initiative architecting high-scale Spring Boot and Postgres microservices in a fintech ecosystem serving 50M+ users.
- Improved app engagement by 70%, managing a team of 4, to implement the SLV module.
- Cut down bounce rate by 18%, collaborated with analytics and product teams to foster dashboards using ELK stack.
- **Skills:** Java, Spring Boot, JUnit, Kafka, Elasticsearch, Kibana, AWS, PostgreSQL, Kubernetes.

LEADERSHIP AND TEACHING

Graduate Teaching Assistant

University of Southern California

September 2024 - December 2024, Los Angeles, CA

- Mentored 70+ graduate students on advanced C++ concepts including GPU rendering, multithreading, ray-tracing, physics systems.
- Strengthened student understanding by creating tutorial documents and debug walkthroughs, conducted weekly office hours.
- **Skills:** C++, CMake, GDB, Python, Maya, Multithreading, GPU.

SKILLS

Languages: Java, Go, C++, Python, JavaScript, C#

Frameworks: Spring Boot, React, Angular, FastAPI, Express, GraphQL

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Elasticsearch

Tools: AWS, GCP, Docker, Kubernetes, Kafka, Git, Jira, ELK

PROJECTS

x86 Operating System | C++, C, CMake, Assembly

- Architected an OS starting from bootstrap process, including kernel threads, distributed design concepts, context switching, copy-on-write.
- Designed fork() system call with page reference tracking, write-fault handling, and VFS layer, demonstrating deep expertise in Unix architecture.

GPU-Accelerated Prime Engine | C, C++, Python, Maya, CMake, OpenCV

github.com/shreyas710/Pool-Ball-Game

- Simulated rigid body dynamics at 60 FPS with 48 ms latency using GPU acceleration and multi-threading.
- Implemented view frustum culling to reduce draw calls by 65%, boosting frame rate by 30% in complex scenes, and developed a Pool Ball game.

Artsy Search | Kotlin, Swift, GCP, JWT, MongoDB

github.com/shreyas710/android-artsy

- Built a cross-platform event search app with responsive design, JWT-based authentication, and MongoDB.
- Configured scalable backend on GCP App Engine; integrated API & media services leveraging Retrofit & Coil.

ACHIEVEMENTS

- Engaged in ICPC 2019 Regionals.
- Secured 800 Rank in Google Hashcode among 1.25 Lakh participants.